WoSM Academy

CS102-F26 · DIGITAL DESIGN AND COMPUTER ARCHITECTURE I

# Boolean Algebra

Lecture 03: the laws and theorems of an algebra of 0 and 1

Dr. Ali Khudiyev

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# Today

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01

# An algebra of 0 and 1

# From school algebra to Boolean algebra

School algebra works with numbers, variables and many operations:

$$3x^2 - \frac{4y}{z} + \frac{2}{7}xz^3, \qquad x, y, z \in \mathbb{R}.$$

In Boolean algebra we restrict both:

- the values to the set $\{0, 1\}$;
- the operations to addition (+), multiplication ($\cdot$) and negation ($'$).

Some laws are the same as in school, some are new. For example, with $x = y = 1$, school algebra gives $x + y = 2$. There is no 2 in $\{0, 1\}$: in Boolean algebra $1 + 1 = 1$.

Named after George Boole, The Laws of Thought (1854); see Lecture 01.

# The three operations

| $a$ | $b$ | $a + b$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

OR

| $a$ | $b$ | $a \cdot b$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

AND

| $a$ | $a'$ |
|---|---|
| 0 | 1 |
| 1 | 0 |

NOT

- $a + b$ is 1 when *at least one* of $a$, $b$ is 1.
- $a \cdot b$ is 1 only when *both* are 1.
- $a'$ is the *complement* of $a$.

As logic gates, these are the OR, AND and NOT gates (Unit 6).

# Notation and precedence

- The complement has three common notations: $a'$, $\neg a$ and $\overline{a}$. These slides use $a'$, as the book does.
- The dot is often left out: $ab$ means $a \cdot b$.
- As in school, multiplication binds more tightly than addition, and the complement binds most tightly of all:

$$a + b'c \;=\; a + \big((b') \cdot c\big).$$

- To complement more than one variable, use brackets: $(ab)'$ is not $a'b'$.

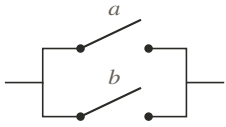
> **Warning**
>
> $(ab)'$ and $a'b'$ are different functions: for $a = 1$, $b = 0$ the first is 1 and the second is 0.

# Switches: series and parallel

Claude Shannon (1937): Boolean algebra describes circuits of switches. Let 1 mean a closed switch, or current flowing.




**Series**: current flows only if $a$ *and* $b$ are closed: $a \cdot b$.




**Parallel**: current flows if $a$ *or* $b$ is closed: $a + b$.

Two switches in parallel, both closed, still carry current: $1 + 1 = 1$.

02

# The seven laws

# Laws 1–3

**Definition (Commutativity)**

The order does not matter: $a + b = b + a, \quad a \cdot b = b \cdot a.$

**Definition (Associativity)**

The grouping does not matter: $(a + b) + c = a + (b + c), \quad (ab)c = a(bc).$

**Definition (Distributivity)**

Expand or factor: $a(b + c) = ab + ac, \quad a + bc = (a + b)(a + c).$

# Laws 4–7

**Definition (Identity)**

0 is the identity for +, and 1 for ·:

$$a + 0 = a, \quad a \cdot 1 = a$$

**Definition (Idempotence)**

An operand combined with itself:

$$a + a = a, \quad a \cdot a = a$$

**Definition (Complement)**

With its complement, the sum is 1 and the product is 0:

$$a + a' = 1, \quad a \cdot a' = 0$$

**Definition (Domination)**

Adding 1 gives 1; multiplying by 0 gives 0:

$$a + 1 = 1, \quad a \cdot 0 = 0$$

# Not your school algebra

- $a + a = a$, not $2a$; $\quad a + 1 = 1$, not $a + 1$.
- The second distributive law, $a + bc = (a + b)(a + c)$, is new. With ordinary numbers it fails: $1 + 2 \cdot 3 = 7$, but $(1 + 2)(1 + 3) = 12$.
- There is no subtraction or division, so you cannot cancel: $a + b = a + c$ does not mean $b = c$. Try $a = 1$, $b = 0$, $c = 1$.

> **Remark**
>
> The laws are the axioms of Boolean algebra: we accept them as they are stated, and everything else is derived from them.

# Duality

Look again at the laws: each one is a pair. The second equation comes from the first by swapping $+ \leftrightarrow \cdot$ and $0 \leftrightarrow 1$.

| Law | Equation | Its dual |
|---|---|---|
| Identity | $a + 0 = a$ | $a \cdot 1 = a$ |
| Complement | $a + a' = 1$ | $a \cdot a' = 0$ |
| Distributivity | $a(b + c) = ab + ac$ | $a + bc = (a + b)(a + c)$ |

> **The duality principle**
>
> If an equation follows from the laws, so does its dual. Every proof gives you a second theorem for free.

# Checking a law by exhaustion

A variable has only two values, so we can try them all. For $a + bc = (a + b)(a + c)$:

| $a$ | $b$ | $c$ | $bc$ | $a + bc$ | $a + b$ | $a + c$ | $(a + b)(a + c)$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | **0** | 0 | 0 | **0** |
| 0 | 0 | 1 | 0 | **0** | 0 | 1 | **0** |
| 0 | 1 | 0 | 0 | **0** | 1 | 0 | **0** |
| 0 | 1 | 1 | 1 | **1** | 1 | 1 | **1** |
| 1 | 0 | 0 | 0 | **1** | 1 | 1 | **1** |
| 1 | 0 | 1 | 0 | **1** | 1 | 1 | **1** |
| 1 | 1 | 0 | 0 | **1** | 1 | 1 | **1** |
| 1 | 1 | 1 | 1 | **1** | 1 | 1 | **1** |

The two bold columns agree on every row, so the equation holds. With $n$ variables there are $2^n$ rows (Lecture 02).

03

# Theorems

# Two ways to prove an equation

### Proof by exhaustion

- Compute both sides for every combination of values.
- Always works, but needs $2^n$ rows: fine for 3 variables, not for 30.

### Algebraic proof

- Turn one side into the other, one law per step.
- Short when you see the way; and each step shows *why*.

**Example (Idempotence follows from the other laws)**

$$
\begin{aligned}
a + a &= (a + a) \cdot 1 && \texttt{identity} \\
&= (a + a)(a + a') && \texttt{complement} \\
&= a + aa' && \texttt{distributivity} \\
&= a + 0 = a && \texttt{complement, identity}
\end{aligned}
$$

# De Morgan's theorem

**Theorem (De Morgan)**

The complement of a sum is the product of the complements, and the complement of a product is the sum of the complements:

$$(a + b)' = a' \cdot b', \qquad (a \cdot b)' = a' + b'.$$

| $a$ | $b$ | $(a+b)'$ | $a' \cdot b'$ | $(a \cdot b)'$ | $a' + b'$ |
|---|---|---|---|---|---|
| 0 | 0 | $0' = 1$ | $1 \cdot 1 = 1$ | $0' = 1$ | $1 + 1 = 1$ |
| 0 | 1 | $1' = 0$ | $1 \cdot 0 = 0$ | $0' = 1$ | $1 + 0 = 1$ |
| 1 | 0 | $1' = 0$ | $0 \cdot 1 = 0$ | $0' = 1$ | $0 + 1 = 1$ |
| 1 | 1 | $1' = 0$ | $0 \cdot 0 = 0$ | $1' = 0$ | $0 + 0 = 0$ |

# Using De Morgan

- It works for any number of variables: $(a + b + c)' = a'b'c'$ and $(abc)' = a' + b' + c'$.
- To complement an expression: swap + and ·, and complement every variable. Keep the original grouping with brackets.

**Example (Complement of $ab + c$)**

$$\begin{aligned}(ab + c)' &= (ab)' \cdot c' && \text{De Morgan} \\ &= (a' + b') \cdot c' && \text{De Morgan}\end{aligned}$$

Coming back: De Morgan's theorem is why a single kind of gate, NAND or NOR, can build every circuit (Unit 6).

# Absorption

**Theorem (Absorption)**

$$a + a \cdot b = a, \qquad a \cdot (a + b) = a.$$

| | |
|---|---|
| $a + ab = a \cdot 1 + a \cdot b$ | identity |
| $= a \cdot (1 + b)$ | distributivity |
| $= a \cdot 1$ | domination |
| $= a$ | identity |

| | |
|---|---|
| $a(a + b) = aa + ab$ | distributivity |
| $= a + ab$ | idempotence |
| $= a$ | the left proof |

In words: if $a$ is 1, both sides are 1; if $a$ is 0, both sides are 0. The value of $b$ never matters.

# Consensus

**Theorem (Consensus)**

$$a \cdot b + a' \cdot c + b \cdot c = a \cdot b + a' \cdot c$$

The term $bc$ is redundant: whenever $b = c = 1$, either $a = 1$ and $ab = 1$, or $a = 0$ and $a'c = 1$.

$$\begin{aligned} ab + a'c + bc &= ab + a'c + bc(a + a') && \texttt{identity, complement} \\ &= ab + abc + a'c + a'bc && \texttt{distributivity, commutativity} \\ &= ab + a'c && \texttt{absorption, twice} \end{aligned}$$

Its dual: $(a + b)(a' + c)(b + c) = (a + b)(a' + c)$.

# Consensus by exhaustion

| $a$ | $b$ | $c$ | $ab$ | $a'c$ | $bc$ | $ab + a'c + bc$ | $ab + a'c$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | **0** | **0** |
| 0 | 0 | 1 | 0 | 1 | 0 | **1** | **1** |
| 0 | 1 | 0 | 0 | 0 | 0 | **0** | **0** |
| 0 | 1 | 1 | 0 | 1 | 1 | **1** | **1** |
| 1 | 0 | 0 | 0 | 0 | 0 | **0** | **0** |
| 1 | 0 | 1 | 0 | 0 | 0 | **0** | **0** |
| 1 | 1 | 0 | 1 | 0 | 0 | **1** | **1** |
| 1 | 1 | 1 | 1 | 0 | 1 | **1** | **1** |

In both rows where $bc = 1$, another term is already 1, so dropping $bc$ changes nothing.

# Transposition

**Theorem (Transposition)**

$$(a+b)\cdot(a'+c) = a\cdot c + a'\cdot b$$

$$\begin{aligned}
(a+b)(a'+c) &= (a+b)\,a' + (a+b)\,c && \text{distributivity}\\
&= a'a + a'b + ac + bc && \text{distributivity, commutativity}\\
&= 0 + a'b + ac + bc && \text{complement}\\
&= ac + a'b + bc && \text{identity, commutativity}\\
&= ac + a'b && \text{consensus}
\end{aligned}$$

The last step is consensus with the roles of $b$ and $c$ swapped. The book shows it by case analysis on $bc$: if $bc = 1$ then $b = c = 1$, and $ac + a'b = a + a' = 1$ as well.

# Involution

**Theorem (Involution)**

$$(a')' = a$$

| $a$ | $(a')'$ |
|---|---|
| 0 | $(0')' = 1' = 0$ |
| 1 | $(1')' = 0' = 1$ |

Complementing twice gives back what you started with: two NOT gates in a row change nothing (Unit 6).

# All in one place

| | | Dual |
|---|---|---|
| Commutativity | $a + b = b + a$ | $ab = ba$ |
| Associativity | $(a + b) + c = a + (b + c)$ | $(ab)c = a(bc)$ |
| Distributivity | $a(b + c) = ab + ac$ | $a + bc = (a + b)(a + c)$ |
| Identity | $a + 0 = a$ | $a \cdot 1 = a$ |
| Idempotence | $a + a = a$ | $aa = a$ |
| Complement | $a + a' = 1$ | $aa' = 0$ |
| Domination | $a + 1 = 1$ | $a \cdot 0 = 0$ |
| De Morgan | $(a + b)' = a'b'$ | $(ab)' = a' + b'$ |
| Absorption | $a + ab = a$ | $a(a + b) = a$ |
| Consensus | $ab + a'c + bc = ab + a'c$ | $(a + b)(a' + c)(b + c) = (a + b)(a' + c)$ |
| Transposition | $(a + b)(a' + c) = ac + a'b$ | |
| Involution | $(a')' = a$ | |

04

# Simplifying expressions

# Two quick ones

$$
\begin{aligned}
ab + ab' &= a(b + b') && \texttt{distributivity} \\
&= a \cdot 1 && \texttt{complement} \\
&= a && \texttt{identity}
\end{aligned}
\qquad
\begin{aligned}
(a + b)(a + b') &= a + bb' && \texttt{distributivity} \\
&= a + 0 && \texttt{complement} \\
&= a && \texttt{identity}
\end{aligned}
$$

The two are duals of each other. Whatever $b$ is, the result depends only on $a$.

> **Coming back**
>
> Combining two terms that differ in one variable, like $ab$ and $ab'$, is the idea behind Karnaugh maps `(Unit 5)`.

# A longer one

Simplify $f = a'b + ab + ab'$.

$$
\begin{aligned}
a'b + ab + ab' &= a'b + a(b + b') && \texttt{distributivity} \\
&= a'b + a \cdot 1 = a'b + a && \texttt{complement, identity} \\
&= a + a'b && \texttt{commutativity} \\
&= (a + a')(a + b) && \texttt{distributivity} \\
&= 1 \cdot (a + b) = a + b && \texttt{complement, identity}
\end{aligned}
$$

Three products and two sums became one sum. Check one row: $a = 0$, $b = 1$ gives $1 + 0 + 0 = 1$ on the left and $0 + 1 = 1$ on the right.

# Why simplify?

- Each operation will become a logic gate (Unit 6). A simpler expression is a smaller, cheaper and often faster circuit.
- $a'b + ab + ab'$ needs two NOTs, three ANDs and two ORs; $a + b$ needs one OR.
- Algebra finds simplifications, but you have to see them. Karnaugh maps give a systematic way for small functions (Unit 5).

# Try it

Simplify, naming the law for each step.

1. $a + a'b$
   Answer: $(a+a')(a+b) = 1 \cdot (a+b) = a+b$ (distributivity, complement, identity)
2. $a(a' + b)$
   Answer: $aa' + ab = 0 + ab = ab$ (distributivity, complement, identity)
3. $(a' + b')'$
   Answer: $(a')'(b')' = ab$ (De Morgan, involution)

> **Remark**
> The first two are a pair of duals; compare them with absorption.

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# Summary

# Summary

- Boolean algebra: the values 0 and 1, and the operations + (OR), · (AND) and $'$ (NOT).
- Seven laws: commutativity, associativity, distributivity, identity, idempotence, complement and domination. Each comes as a dual pair.
- Theorems: De Morgan, absorption, consensus, transposition and involution.
- Prove equations by exhaustion ($2^n$ rows) or algebraically, one law per step.
- Simplifying an expression simplifies the circuit that computes it.

# Before the next lecture

- Read Section 2.2 of the book, *Boolean Algebra*.
- Prove the dual of absorption, $a(a+b) = a$, by exhaustion.
- Prove the dual of consensus, $(a+b)(a'+c)(b+c) = (a+b)(a'+c)$, by the duality principle and by exhaustion.
- Simplify $abc + abc' + ab'$.

**Next lecture** (Unit 4)

Boolean functions: truth tables, and the canonical sum-of-products and product-of-sums forms.